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Study on Low-Frequency Bandgap Characteristics of Double-Opening Helmholtz Local Resonance Periodic Structures (Jiang Jiulong, Yao Hong, Du Jun, Zhao Jingbo, Deng Tao)

Low frequency band gap characteristics of double-split Helmholtz locally resonant periodic structures

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Study on low-frequency bandgap characteristics **of double-opening Helmholtz local resonant periodic structures** ÿ

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A double-opening Helmholtz periodic structure was designed. This periodic structure unit employs a double-opening inner and outer cavity design. Based on the multi-cavity coupled local resonance mechanism, it can significantly increase the local resonance region and achieve a lower low-frequency bandgap. By adjusting the inner cavity arc length, the bandgap can be shifted to achieve sound insulation effects in a specific low-frequency band. In analyzing the formation mechanism and influencing factors of the low-frequency bandgap, a mathematical model for calculating the bandgap's start and cutoff frequencies was established using the acoustic-electric analogy principle and compared with the finite element method. The study shows that this structure has good low-frequency bandgap characteristics, with its lowest bandgap range being 86.9–138.2 Hz. Under a fixed outer diameter, the low-frequency bandgap is affected by the inner cavity arc length, the spacing between the inner and outer cavities, and the spacing of the periodic unit structure. The longer the inner cavity arc length, the lower the low-frequency bandgap; the larger the distance between the inner and outer cavities, the smaller the inner cavity volume, causing the bandgap to shift towards higher frequencies and deteriorating the low-frequency performance; reducing the structural spacing has no effect on the low-frequency bandgap start frequency. However, it can significantly increase the low-frequency bandgap cutoff frequency, thereby increasing the bandgap width. The findings of this study can provide some practical and theoretical support for the field of low-frequency noise reduction.

Keywords: Helmholtz resonance, periodic structure, low-frequency bandgap, acoustic-electric analog

PACS: 43.20.+g, 42.50.Gy, 46.40.Ff **DOI:** 10.7498/aps.66.064301

1 Introduction

In recent years, artificial periodic structures, represented by phononic crystals and acoustic metamaterials, have attracted increasing attention from researchers. In artificial periodic structures, the periodically changing structural parameters give them a "bandgap" characteristic. This bandgap characteristic manifests as the inability of sound waves or elastic waves with frequencies within the bandgap range to propagate within the structure. This characteristic lays the foundation for the application of such structures or materials in vibration isolation and noise reduction. Long-term research indicates that the generation of bandgap in artificial periodic structures mainly occurs in two ways, the other being localized... Mechanisms: One is the Bragg scattering mechanism [1ÿ3] and the other is the resonance mechanism [4,5]. The band gap in a Bragg-scattering phononic crystal is called the Bragg band gap, and its formation is controlled by the Bragg condition—the lattice size must be at least approximately half the wavelength of the elastic wave. Therefore, to obtain a low-frequency Bragg band gap, the required phononic crystal size tends to be... The size is too large, making it inconvenient for practical applications. The local resonance mechanism was first proposed by Liu et al. [4] . They found that this new type of phononic crystal can generate very low frequency band gaps under small size conditions (the lattice size is much smaller than the elastic wave wavelength corresponding to the band gap frequency), which greatly breaks through the traditional The limitations of Bragg scattering phononic crystals have been overcome, thus achieving the goal of "small size controlling large wavelength". Local resonant phononic crystals can also exhibit novel properties such as negative equivalent mass density and negative equivalent modulus [6-12] . Currently, the main development direction of using artificial periodic structures to deal with vibration isolation and noise reduction problems is to design "lightweight, compact, low-frequency broadband" structures. Helmholtz resonant cavities utilize the resonance characteristics of air, and only require structural design without the need for additional mass, which gives them good advantages in lightweight and low-frequency design, and has been recognized by many scholars. Fang et al. [13] used the Helmholtz resonance principle to design a periodic structure, which for the first time realized the structure in a

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It has a negative Young's modulus. Ding et al. [14,15] proposed a controllable two-dimensional Helmholtz periodic structure, and theoretical analysis and experiments showed that it also has a negative Young's modulus. Guan et al. [16] designed a Helmholtz resonator with a cycloid structure and obtained a good low-frequency bandgap. Li et al. [17] designed a multi-layer open ring structure, opened a lower low-frequency bandgap, and used the acoustic-electric analogy method to analyze the parameters affecting its bandgap. Liu Min et al. [18] pointed out that there is a weak coupling effect between the resonant unit and the matrix and the coupling effect between the resonant units in the local resonant phononic crystal through the study of the acoustic characteristics of the two-dimensional square cylindrical Helmholtz resonant cavity array. Murray et al. [19] designed a double fishnet structure based on the Helmholtz resonance principle and obtained a good sound insulation effect in the mid-to-high frequency range.

Current research on the periodic structure of Helmholtz resonators rarely considers low-frequency characteristics, especially in the frequency range below 100 Hz. This is mainly because the local resonance region in the periodic structure design of Helmholtz resonators is relatively small due to the influence of the number and depth of openings, often requiring a larger volume to obtain low-frequency characteristics. In this paper, the periodic structure unit adopts a double-opening inner and outer cavity design. The double-opening design increases the area of communication between the resonator and the external air; the unique inner and outer cavity design increases the opening depth to enlarge the synergistic resonance region, thus lowering the resonance frequency. Near the resonance frequency, because the air is localized inside the structure, sound waves cannot propagate, achieving sound insulation. This manifests as a downward shift in the band structure, resulting in lower low-frequency band characteristics. Simultaneously, by adjusting the arc length of the inner cavity, the bandgap frequency band can be shifted, achieving adjustable low-frequency bandgap. In analyzing the formation mechanism of the low-frequency bandgap, an equivalent mathematical model of the low-frequency bandgap circuit is established and compared with the finite element method. This mathematical model significantly reduces the difficulty of structural design, saves a considerable amount of finite element calculation time, and allows for complete... Quantitative analysis of the influence of structural parameters on bandgap can reveal in depth the impact of structural parameters on low-frequency bandgap.

2. Low-frequency bandgap of a double-aperture localized resonant phonon crystal

The cross-section of the double-opening local resonant unit structure is shown in Figure 1(a). This structure is a cylindrical structure with inner and outer cavities, made of steel. The outer cavity radius is r; the opening widths are w1 and w2 respectively ; the tube wall thickness is d; the inner cavity arc angle is ÿ; and the interval between the two cavities is w0. In the structural design, the structure can be changed by

adjusting the inner cavity arc angle ÿ, achieving adjustable band gap. To investigate the band structure and resonant modes of this structure, a model with structural parameters a = 100 mm, r = 45 mm, w0 = 1 mm, d = 1 mm, w1 = 2 mm, w2 = 2 mm, and ÿ = 180° was taken as the research object. For the air domain, the frequency domain expression of the sound pressure Helmholtz equation is:

$$\ddot{y}\left(\frac{1}{\dot{y}}\dot{y}p\right) + \frac{2\dot{y}}{\dot{y}c^2}p = 0, \tag{1}$$

In the formula, p is the sound pressure, ÿ is the angular frequency, c is the sound velocity, and ÿ is the

air density. The acoustic impedance of air is much smaller than that of solid steel. Due to the large difference and mismatch in acoustic impedance between the two media, sound waves have difficulty propagating from one medium to the other. Sound waves propagating in air are almost entirely reflected between solids and rarely penetrate. It can be assumed that the energy of sound waves passing through solid steel is very small, and the energy that can cause solid vibration is extremely small, almost negligible. Therefore, in the calculation, the cavity shell is considered a rigid body, and its vibration is not considered. According to Bloch theory, the Bloch-Floquet boundary is used in this structure, and its expression is:

$$p(r + a) = p(r)\exp(ika), \tag{2}$$

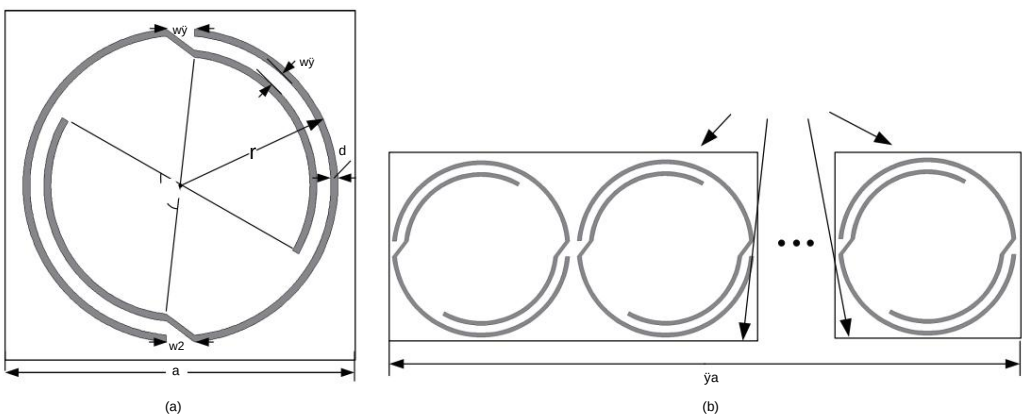

Fig. 1. (a) The cross section of the double-split Helmholtz resonant structure; (b) The finite structure with 5 cells of the double-split Helmholtz resonant structure.

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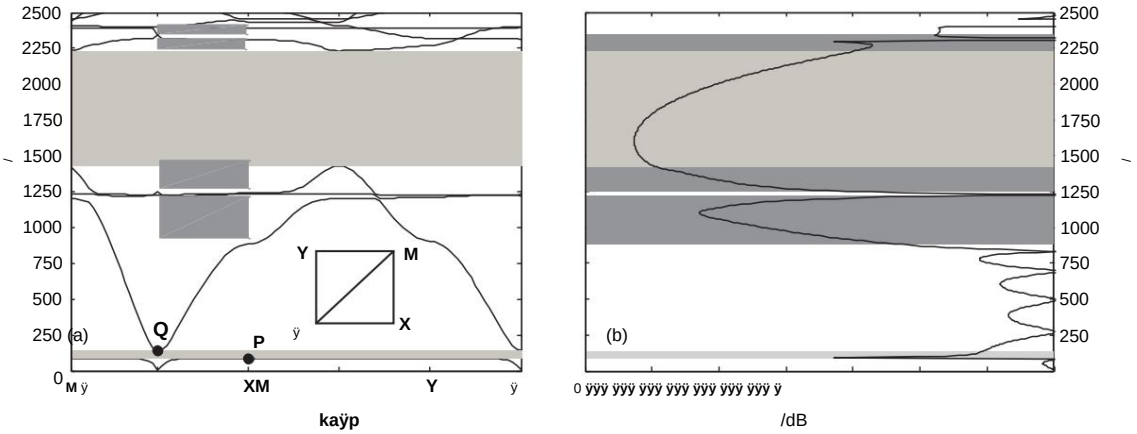

Fig. 2. (a) Band diagram of the double-split Helmholtz resonant structure; (b) The transmission spectra of the double-split Helmholtz resonant structure.

In the formula, r is the position vector; a is the lattice constant of the phonon crystal; and parameter k is the wave vector, which describes the relationship between the phase and the defined boundary conditions and the cells. Under the given wave vector conditions, a set of eigenvalues and eigenvectors can be obtained by solving the spectral equations. The eigenvector corresponding to the wave vector on the boundary of the first Brillouin region represents the sound

pressure of the characteristic mode. To calculate the acoustic transmission characteristics of the proposed structure, a periodic structure consisting of 5 finite number of cells is defined. The structure is infinitely periodic in the y-direction and consists of 5 cells in the x-direction, as shown in Figure 1(b). Periodic boundary conditions are applied to the top and bottom edges of the structure, while perfectly matched layers are applied to both ends.

For a finite phonon crystal structure, the sound insulation (or transmission loss) can be calculated using the following formula:

$$T = \ddot{y}20 \lg \frac{|Po|}{|Pi|}, \qquad (3)$$

In the formula, Po is the output sound pressure and Pi is the incident sound pressure. By changing the plane wave transmission frequency, the sound insulation curve is obtained after repeated calculations. The calculated bandgap structure and sound insulation are shown in Figure 2(b).

The COMSOL Multiphysics simulation software (acoustic module) was used, and the default triangular mesh generation in the software module was employed to calculate the bandgap structure and transmission spectrum. The results are shown in Figure 2. It can be seen that in the frequency band below 2500 Hz, the structure has two complete bandgap segments, as shown in the light gray area of Figure 2 (a). The lowest bandgap is located between the first and second energy bands, with a bandgap width of 87.1–138.2 Hz. The second complete bandgap is 1428.5–2231.7 Hz. It is also noted that this structure has four directional band gaps in the ÿ-X direction, as shown in the dark gray area of Figure 2 (a). To verify the correctness of the band gap structure calculation, a sound insulation characteristic diagram of a 5-cell finite structure is also given for comparison, as shown in Figure 2 (b).

As mentioned earlier, wave propagation within the bandgap is suppressed. The sound insulation characteristic diagram shows that sound waves are significantly suppressed across the entire bandgap and in the ÿ-X direction bandgap frequency band, and the suppression range matches the bandgap width well, demonstrating the correctness of the bandgap calculation.

Further analysis reveals that the first band exhibits a relatively long flat state in most of the Brillouin region, indicating the existence of local resonant modes in the structure, and its first bandgap is a local resonant bandgap. The second complete bandgap is a Bragg bandgap, and its frequency position is mainly controlled by the Bragg condition. The Bragg condition states that to realize a Bragg bandgap in a phononic crystal, its lattice size must be at least approximately half the wavelength of the elastic wave. Its bandgap frequency center can be obtained from the following equation:

$$f = 2a \frac{nc}{} \quad (n = 1, 2, 3 \ldots \cdot \cdot \cdot), \qquad (4)$$

Where c represents the elastic wave velocity of the matrix in the phononic crystal, and a represents the lattice constant. Therefore, in a Bragg scattering phononic crystal, the center frequency of its first bandgap is generally located near c/(2a). Thus, the first bandgap frequency of this structure... The Bragg bandgap center frequency is calculated based on the lattice constant and the speed of sound in air, i.e., f0 = c0/(2a) = 1715 Hz, which is consistent with the band structure diagram 2 (a). At the same time, it can also be seen from equation (4) that the formation of the second bandgap is mainly affected by the lattice constant and has little to do with the structural design. The cell structure mainly affects the low-frequency bandgap. Therefore, in the following text, we will mainly analyze the influence of the structure on the formation mechanism of the low-frequency bandgap.

3. Low-frequency bandgap formation mechanism and equivalent modeling analysis

As can be seen from the bandgap diagram in Figure 2, the first band exhibits a relatively long flat state in most of the Brillouin region, indicating the existence of local resonant modes in this structure. To further investigate the structure's low-voltage band...

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To investigate the bandgap formation mechanism, the sound pressure fields at the upper and lower limits of the low-frequency full bandgap (points P and Q in Figure 2 (a)) were selected to analyze the formation mechanism. Figure 3 (a) shows the sound pressure field of the P mode. It can be seen that the sound pressure is almost entirely confined inside the circular cavity, while the external sound pressure field is very small, almost zero. However, within the annular channel formed by the inner and outer cavities, the sound pressure field varies, being almost zero on the outside and increasing closer to the inner circular cavity. Therefore, it can be considered that the P mode is mainly due to the resonance of sound waves within the structure, with the sound wave energy confined to the structure.

The P mode cannot propagate externally. In fact, the P mode determines the lower limit of the low-frequency bandgap. The Q mode, however, is the complete opposite of the P mode. As shown in Figure 3 (b), the sound pressure is entirely concentrated outside the structure, while the sound pressure inside the structure is almost zero. Within the annular channel, the sound pressure variation is also opposite to that of the P mode, being high outside and low inside. This indicates that the sound wave propagates outside the structure, and the structure has no effect on the propagation of the sound wave. Therefore, the Q mode determines the cutoff point of the bandgap, which is the upper limit of the low-frequency bandgap. A mathematical model can be established using the acoustic-electric analogy method to further analyze the influence of the two modes on the bandgap.

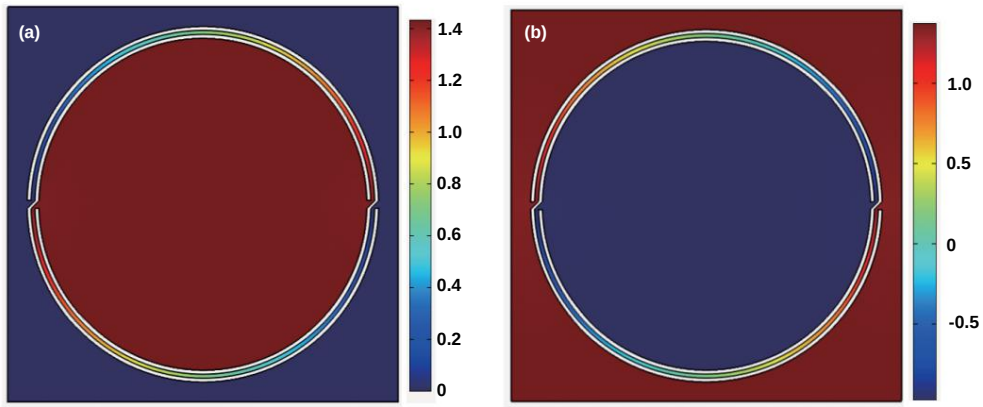

Fig. 3 (color online) (a) Sound pressure distribution diagrams of point P; (b) Sound pressure distribution diagrams of point Q.

For ease of description, in the process of analyzing the bandgap formation mechanism of the double-opening Helmholtz resonance periodic structure, the structure is divided into four parts, as shown in Figure 4. In this division, A represents the outer air layer of the structure; B and C are the annular channels formed by the inner and outer cavities; and D is the circular inner cavity.

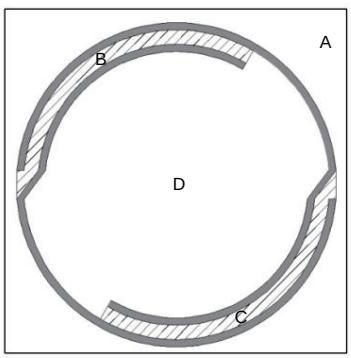

Figure 4 Schematic diagram of structural region division

Fig. 4. The region division of the cross section in structure.

For mode P, since the external acoustic pressure field is almost zero, mode P is mainly generated by the coordinated resonance of regions B, C, and D, and is independent of region A; therefore, region A is not considered. According to acoustic-electric...

By analogy, the narrow, annular channels B and C can be equivalent to inductors. LB, LC, and the circular inner cavity is equivalent to a capacitor CD, and the circuit formed by them is shown in Figure 5.

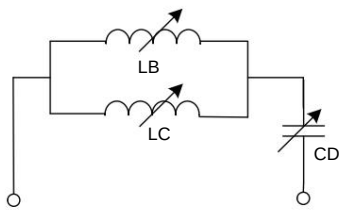

Figure 5. Equivalent circuit diagram of mode P

Fig. 5. The equivalent circuit of point P.

In Figure 5 , LB and LC are the equivalent inductances, and their expressions are:

$$LB = LC = \frac{\ddot{y}\,[\ddot{y}\,(r\,\ddot{y}\,d\,\ddot{y}\quad {}^{\frac{1}{2}} w0\,)]}{w0h}, \tag{5}$$

$$CD = \frac{[\ddot{y}(r\,\ddot{y}\,d)\quad {}^{2}\,\ddot{y}\,\ddot{y}(d + w0)(2r\,\ddot{y}\,3d\,\ddot{y}\,w0)]h\,\ddot{y}c2}{}. \tag{6}$$

The formula for calculating the resonant frequency of mode P is as follows:

$$fP = \frac{1}{2\ddot{y}\ddot{y}LCCD}. \tag{7}$$

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For mode Q, since regions A, B, C, and D all possess sound pressure fields, all four regions must be fully considered in the acoustic-electric analogy of mode Q. Region A is equivalent to a capacitor, and the resulting resonant circuit is shown

Figure 6. (a CA = $\frac{\text{in}^2 \; 2\ddot{y}\,\ddot{y}r)h}{\ddot{y}c2}$, (8)

$$fQ = \frac{1}{2\ddot{y}\,\ddot{y}\,\sqrt{LC\,\frac{CDCA}{CD + CA}}} = \frac{\ddot{y}\,\sqrt{CD + CA}}{\ddot{y}\,\ddot{y}\,\sqrt{2LCCDCA}}. \quad (9)$$

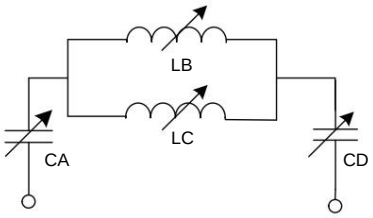

Figure 6. Equivalent circuit diagram of modal Q

Fig. 6. The equivalent circuit of point Q.

4. Analysis of factors affecting low-frequency bandgap

The above establishes a mathematical model for the upper and lower limits of the low-frequency bandgap of the double-opening Helmholtz resonator. This model will help us analyze the factors affecting the low-frequency bandgap. According to equations (5)-(9), the main factors affecting the low-frequency bandgap of this structure are: the inner cavity arc angle ÿ, the interval w0 between the rings , and the lattice constant a, which is also the periodic arrangement interval a ÿ 2r. Among them, the inner cavity arc angle ÿ is the main factor. The change of the inner cavity arc angle will cause a large change in the structure, resulting in changes in the length of the annular channel formed by the inner and outer cavities and the volume of the inner cavity, thereby changing the co-resonance region and affecting the equivalent parameters of the Helmholtz resonator. Below, we will analyze these three influencing factors

separately. In order to control variables for easy comparison and analysis, when analyzing the influence of the inner cavity arc angle ÿ of the double-opening Helmholtz resonator on the low-frequency bandgap, we set the interval w0 between the rings to 1 mm and the lattice constant a to 100 mm. When the inner cavity arc angle ÿ increases from 15ÿ to 180ÿ , The upper and lower limits of the bandgap were calculated using the established bandgap acoustic-electric analogy mathematical model and finite element software, and the results were compared and verified to obtain the influence of the cavity arc angle ÿ on the low-frequency bandgap . The results are shown in Figure 7.

In Figure 7 , the red, blue, and green dashed lines represent the upper and lower limits and the bandgap width of the first low-frequency bandgap calculated using the acoustic-electric analogy mathematical model, respectively. The inverted triangle, triangle, and circle represent the upper and lower limits and the bandgap width of the first low-frequency bandgap calculated using the finite element method (this is the case in all subsequent figures and will not be repeated). From Figure 7...

It can be seen that the error between the results calculated using acoustic-electric analogy and those calculated using finite element method is small, indicating the correctness of the established model. With the increase of ÿ, the upper and lower limits of the first low-frequency bandgap continuously decrease. The increase of the cavity arc angle ÿ actually increases the length of the inter-ring channel, greatly increasing the equivalent inductance, while also reducing the area of the final cavity. However, the reduction in area is relatively small, resulting in a smaller change in the equivalent capacitance. The lower limit of the first bandgap is only related to regions B, C, and D. The increase in the equivalent inductance of regions B and C and the slight decrease in the equivalent capacitance of region D ultimately cause the lower limit of the first bandgap to continuously decrease, resulting in better low-frequency characteristics. For the first bandgap, its upper limit is the result of the coordinated resonance of the four regions, with an equivalent capacitance of CACD/ (CA + CD). It can be seen that with the increase of the equivalent inductance, its equivalent capacitance further decreases. However, since the capacitance of region A remains unchanged, and the equivalent capacitance of region A is smaller than the internal cavity capacitance, the decrease in its equivalent capacitance is not significant, and is smaller than that due to the increase in the cavity arc angle ÿ. The increase in length of the annular channel leads to an increase in equivalent inductance, thus reducing the upper bandgap. By adjusting the inner cavity arc angle ÿ, the bandgap can be shifted to achieve sound insulation in a specific frequency band.

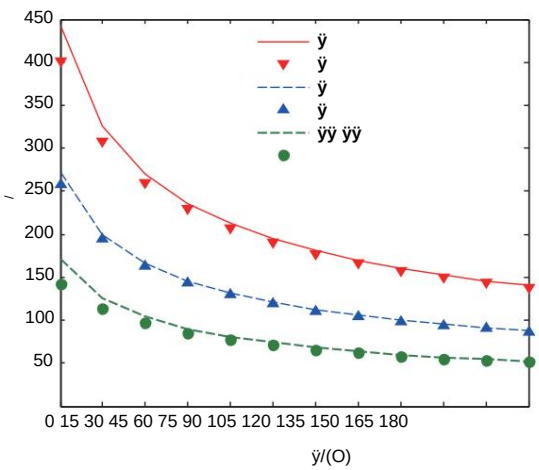

Figure 7 (color online) shows the effect of the cavity arc angle ÿ on the first low-frequency band gap.

When analyzing the effect of the interring spacing w0 on the low-frequency bandgap of a double-opening Helmholtz resonator , the lattice constant a = 100 mm and the cavity arc angle ÿ = 180° were set . When the interring spacing w0 increased from 1 mm to 4 mm, the upper and lower limits of the bandgap were calculated using the established bandgap acoustic-electric analogy mathematical model and finite element software, and the results were compared and verified. The effect of the interring spacing w0 on the low-

frequency bandgap is shown in Figure 8. Figure 8 shows that the error between the acoustic-electric analogy calculation and the finite element calculation is small in the first half, but the error increases significantly when w0 increases from 2 mm to 4 mm. The main reason for

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As the inter-ring spacing increases, the annular channel formed between the rings is no longer a narrow pipe, thus losing the premise of the Helmholtz acoustic-electric analogy and leading to increased calculation errors. However, the overall trend remains consistent with the actual values. It is easy to see that as the inter-ring spacing continues to increase, with little change in the equivalent inductance, the area of the internal cavity decreases significantly, thereby greatly reducing the equivalent capacitance. This results in an increasing trend in both the upper and lower limits of the first low-frequency bandgap. The bandgap shifts significantly towards higher frequencies, and while the bandgap width increases, its low-frequency characteristics are greatly reduced.

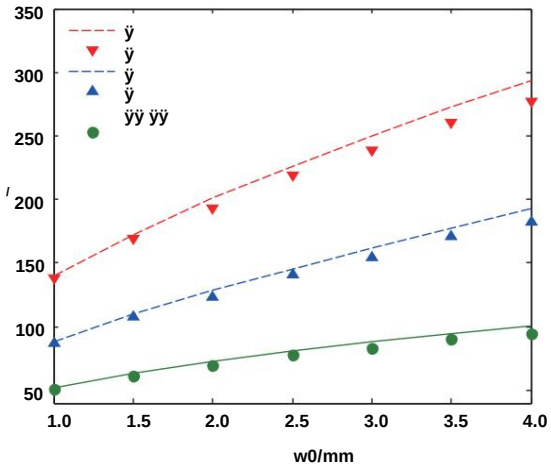

Figure 8 (color online) shows the impact of the inter-ring spacing w0 on the first low-frequency band gap .

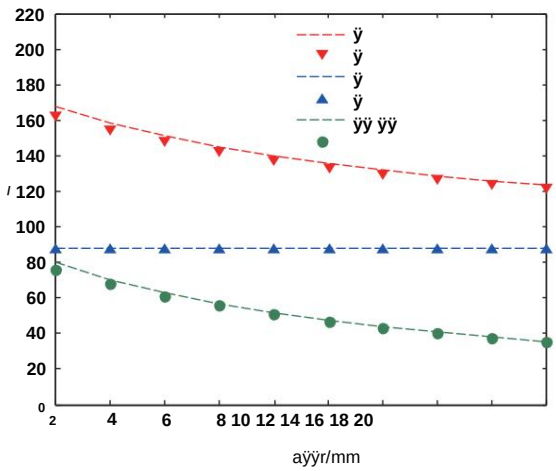

Figure 9 (color online) shows the impact of the interval between each ring on the first low-frequency band gap.

When analyzing the effect of the structural periodic arrangement interval a ÿ 2r of a double-opening Helmholtz resonator on the low-frequency bandgap, the inter-ring interval w0 = 1 mm and the inner cavity arc angle ÿ = 180° were set. When the structural periodic arrangement interval a ÿ 2r increased from 2 mm to 20 mm, the established bandgap acoustic-electric analogy mathematical model and finite element software were used to analyze the bandgap. The upper and lower limits of the gap were calculated and compared to verify the results. The influence of the periodic arrangement interval a ÿ 2r on the low-frequency bandgap is shown in Figure 9.

Figure 9 illustrates the effect of the periodic arrangement spacing on the bandgap structure. It can be seen that the periodic arrangement spacing has virtually no effect on the lower bandgap limit, but a significant impact on the upper bandgap limit. Specifically, as the arrangement spacing increases, the upper bandgap limit continuously decreases, and the bandgap width also decreases accordingly. From the equivalent circuit model, it can be seen that a decrease in the arrangement spacing, i.e., a decrease in the lattice constant 'a', leads to a decrease in the equivalent capacitance of region A. This, in turn, reduces the capacitance after series connection with the equivalent capacitance within the cavity, resulting in an increase in frequency. When the arrangement spacing is 2 mm, the upper and lower bandgap limits are respectively... The bandgap range is greatly increased to 87.2 Hz and 162.7 Hz. On the other hand, the gaps between structures are reduced, that is, the fill rate is increased, and the bandgap width can be widened considerably.

5. Conclusion

In addressing low-frequency noise reduction, conventional methods, due to the mass density theorem, result in significant structural mass and volume. Therefore, "low-frequency, lightweight, and broadband" becomes the desired approach. This paper establishes a double-opening Helmholtz periodic structure that requires no additional mass, laying the foundation for solving the lightweight problem. The unique inner and outer cavity design of the double-opening Helmholtz periodic structure increases the opening depth, thereby expanding the resonant region and achieving lower low-frequency bandgap characteristics. Adjusting the inner cavity arc length allows for shifting the bandgap frequency band, achieving adjustable low-frequency bandgap. Increasing the inner cavity arc length, decreasing the spacing between the inner and outer cavities, and decreasing the structural unit spacing can significantly widen the low-frequency bandgap, achieving broadband low-frequency performance. These findings will benefit the application of this structure in sound insulation and noise reduction, especially in low-frequency noise reduction.

References

[1] Lai Y, Zhang X, Zhang ZQ 2001 *Appl. Phys. Lett.* **79** 3224

[2] Hussein MI, Hulbert GM, Scott RA 2006 *J. Sound Vib.* **289** 779

[3] Hussein MI, Hulbert GM, Scott RA 2007 *J. Sound Vib.* **307** 865

[4] Chen Y, Li J, Zhou J, Huang T, Zhou M, Yu DY 2014 *Shock Vib.* **2014** 189539

[5] Liu ZY, Zhang XX, Mao YW 2000 *Science* **289** 1734 [6] Liu Z, Chan CT, Sheng P 2005 *Phys. Rev. B* **71** 014103 [7] Li J, Chan CT 2004 *Phys. Rev. E* **70** 055602 [8] Deng K, Ding Y, He Z, Zhao H, Shi J, Liu Z 2009 *J. Appl. Phys.* **105** 124909

[9] Maldovan M 2013 *Nature* **503** 209

[10] Zhang S, Yin L, Fang N 2009 *Phys. Rev. Lett.* **102** 194301

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[11] Mei J, Ma G, Yang M, Yang Z, Wen W, Sheng P 2012 *Nat. Commun.* **32** 756

[12] Zhou X, Badreddine Assouar M, Oudich M 2014 *J. Appl. Phys.* **116** 194501

[13] Fang N, Xi D, Xu J, Ambati M, Srituravanich W, Sun C 2006 *Nat. Mater.* **5** 452

[14] Ding C, Hao L, Zhao X 2010 *J. Appl. Phys.* **108** 074911 [15] Hao LM, Ding CL, Zhao XP 2012 *Appl. Phys. A* **106** 807

[16] Guan D, Wu JH, Li J, Gao NS, Hu M 2015 *Noise Control Eng. J.* **63** 20 [17] Li J, Wu JH, Guan D, Gao NS 2014 *J. Appl. Phys.* **116** 103514

[18] Liu M, Hu ZL, Fu XJ 2012 *Acta Phys. Sin.* **61** 104302 (in Chinese) [Liu Min, Hou Zhilin, Fu Xiujun, Acta Physica Sinica, 2012, **61 ,** 104302]

[19] Murray ARJ, Summers IR, Sambles JR, Hibbins A P 2014 *J. Acoust. Soc. Am.* **136** 980

Low frequency band gap characteristics of double-split Helmholtz locally resonant periodic structures*

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Abstract

A double-split Helmholtz periodic structure with the characteristic of local resonance is designed and constructed in this paper. The double-split periodic structural cell which can be divided into internal and external cavities is adopted in structure. In such a kind of structure, the resonating area is remarkably expanded while the inner cavity is continuously enlarged. Thus, a satisfactory feature of low frequency resonance can be obtained. At the same time, the adjustability of band gap is achieved by the designed adjustment of the arc length of the inner cavity, therefore, the Effect of sound insulation in a specific low frequency band can be achieved. In the analyzes of the mechanism and factors of the generation of low frequency band gap, the mathematical model of the upper and lower limits of the band gap is established by using the electric circuit analogy. And some comparative analyzes between the methods of electric circuit analogy and finite element method are carried out. frequency band gap can be influenced by the arc length of inner cavity, the space between inner and outer cavities, and the interaction of the structural cells in the periodic arrangement. Plenty of practical and theoretical support in the field of low frequency noise reduction is offered in the research.

Keywords: Helmholtz resonance, periodic structure, low frequency band gap, electric circuit analogy

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